Pitchblade: Vocal Tuning and Processing Multitool Plugin

User Manual

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1. Introduction

Welcome to Pitchblade, a modular vocal multi-tool audio plugin. Pitchblade is designed to provide comprehensive vocal processing in a single interface. With this tool, you are able to combine and reorder various effects to suit your needs.

Key features include:

- Real-time pitch correction and shifting
- Modular daisy chain workflow allowing you to reorder effects on the fly
- Visual feedback with dedicated real-time graphs for every effect

2. Installation and Setup

Pitchblade runs as a VST3 plugin or a standalone application.

2.1 Prerequisites

- Windows 10/11 (64 bit)
- A VST3-compatible Digital Audio Workstation (DAW) (for VST3 usage)

2.2 Compilation

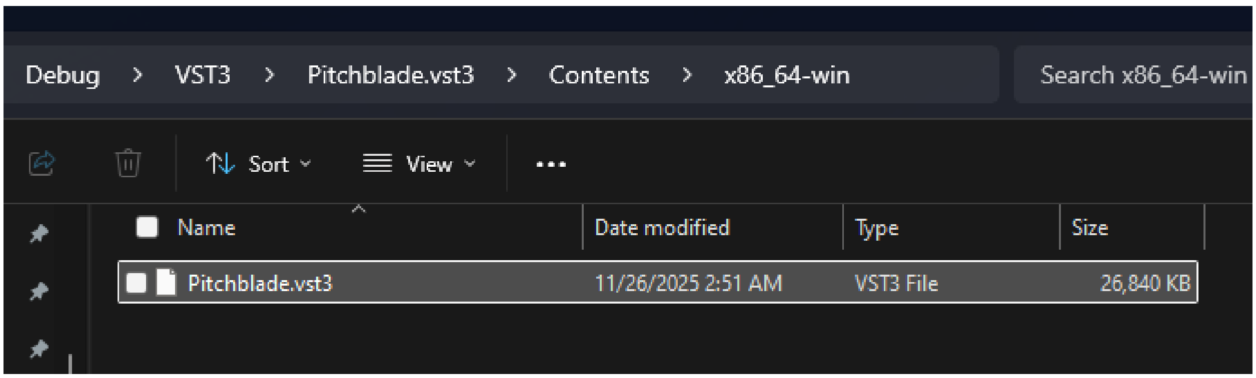
If you want to build this plugin from source, take the following steps:

1. Ensure you have CMake, a C++ compiler, Git, and NSIS installed as described in the readme.
2. Clone the repository for Pitchblade onto your local device.
3. In the root folder for Pitchblade, run .\configure_windows_release.bat for the release version.
    a. You can also run .\configure_windows.bat to compile the debug version, but it does not come with an installer *.exe file.

2.3 Installation

If you want to use the VST3 version of the plugin, take the following steps:

1. Locate the Pitchblade.vst3 file in the build artifacts folder (C:\Users\arh03\Pitchblade\build\plugin\Pitchblade_artefacts).
2. Copy this file to your system's VST3 folder (commonly C:\Program Files\Common Files\VST3).
3. Rescan plugins within your DAW.

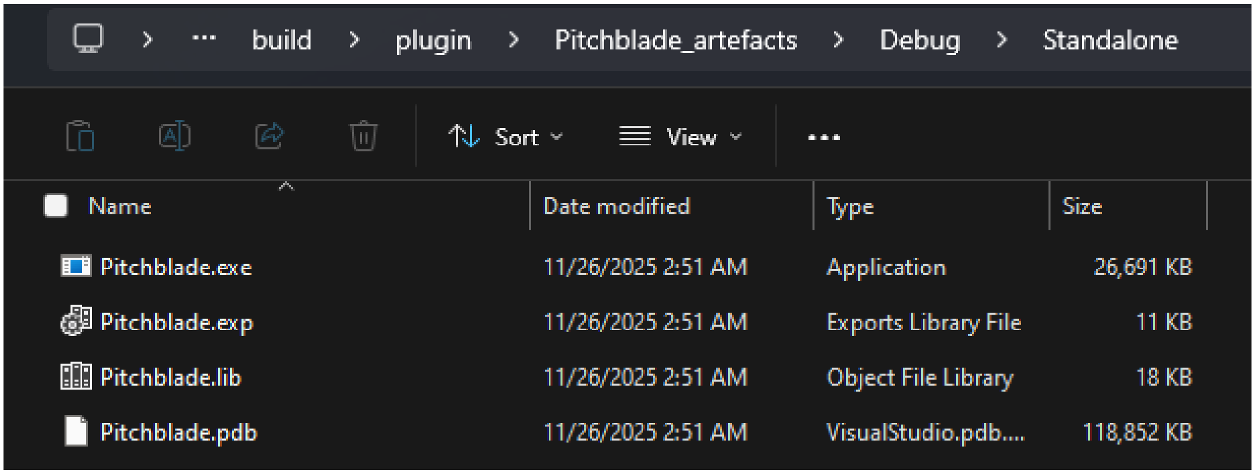


Alternatively, you can use the Pitchblade_Installer.exe to handle these steps for you, in which case you will take the following steps:

1. Run Pitchblade_Installer.exe.
2. Click through the installation window.
3. Rescan plugins within your DAW.

If you want to use the standalone version instead of the VST3 version, then take the following steps:

1. Locate the Pitchblade.exe file in the build artifacts folder (Pitchblade\build\plugin\Pitchblade_artefacts).
2. Run the file.



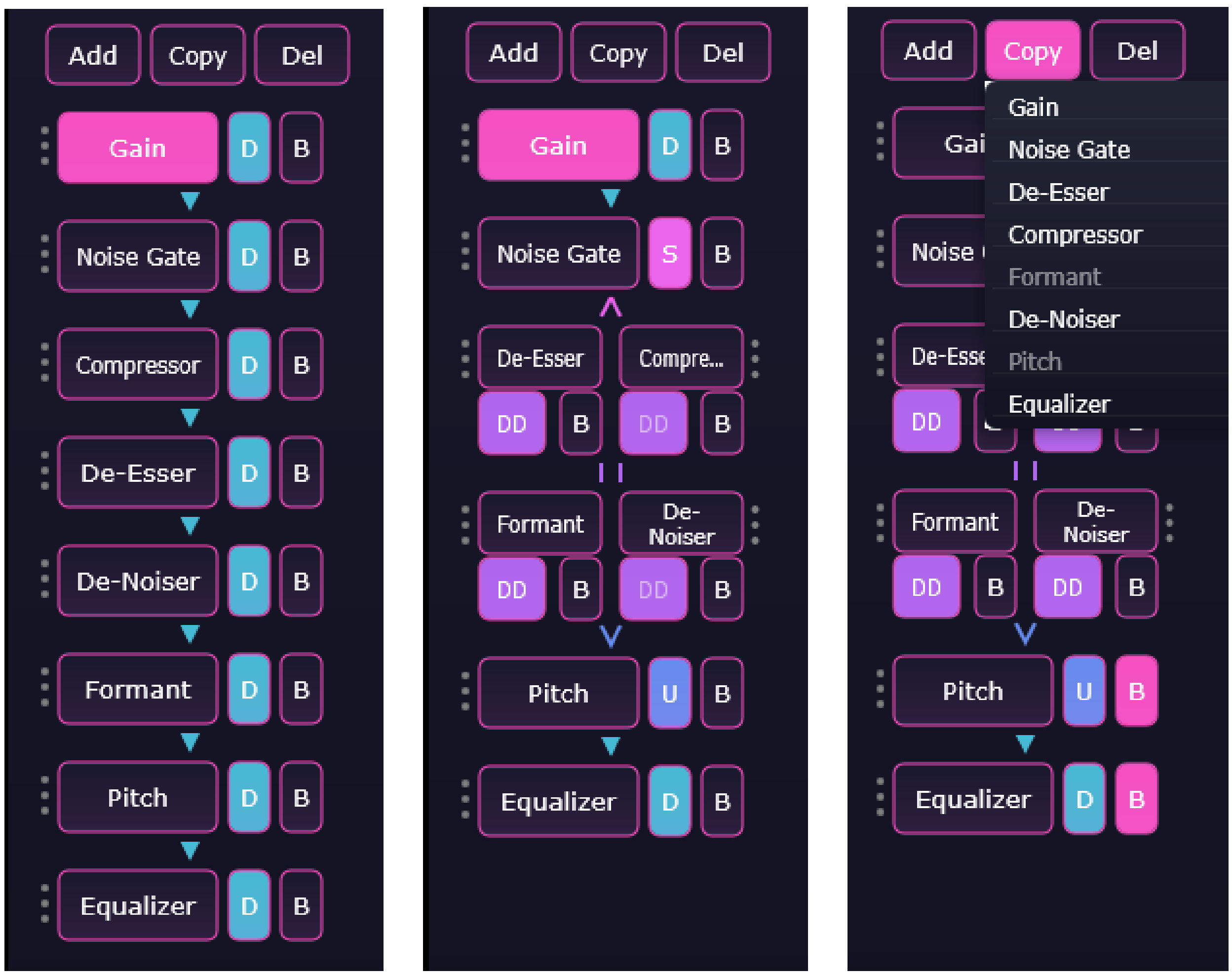
3. Interface Overview

The Pitchblade interface is divided into three main sections. This layout makes it easy to manage large vocal chains without losing track of effects:

1. The top bar - Contains various global options that affect the entire plugin.
2. The daisy chain - Displays the signal path and allows for navigation between effects.
3. The panel view - Contains the effect module parameters and visualizers.

4. The Daisy Chain

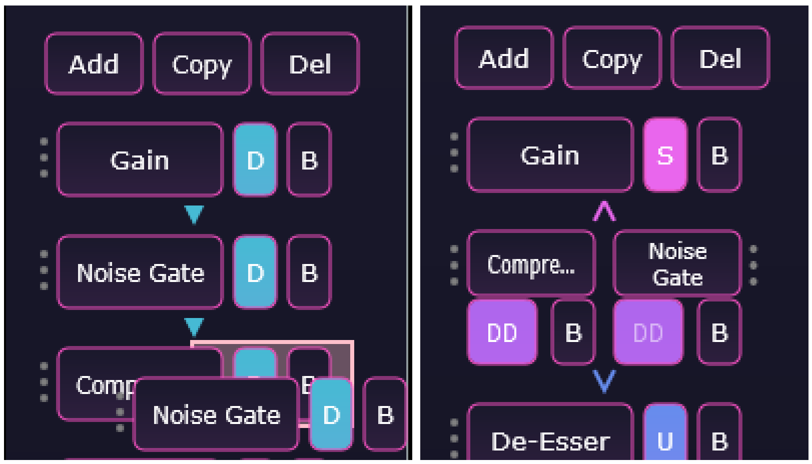
One of Pitchblade's most powerful features is the daisy chain, which allows you to customize your signal path. The chain lists every active effect top-to-bottom. Effects can be reordered, merged on the same row, added, duplicated, or removed. Users can drag effects to change the chaining mode, or how the signal routes together.



4.1 How to Operate

- **Navigation** - Click on any effect name in the daisy chain to open that effect's control panel and visualizer in the panel view.
- **Bypass** - Click "B" on any effects row, and that effect will "turn off". Clicking it again will "turn on" the effect again.

- **Reordering** - Click and drag an effects handle to move it up or down the chain. Changing the order in how effects are stacked together will change their output.
    - For example, placing a Gain module before a Compressor module will result in a different sound than placing it after.
- **Merging Rows** - Click and drag an effects handle onto another effect will add the effect to the same row, merging it together. All effects surrounding it will automatically have their Chaining Mode updated.



- **Chaining Mode** - Displayed on the badge beside each effect's name. Chaining mode describes how all the effects are chained together into one signal path. Just like reordering, how the audio is chained together will change its output.
    - **“D” Down** - Normal one-to-one routing.
    - **“S” Split** - Sends one effect’s output into two parallel effects.
    - **“DD” Double Down** - Processes two effects in parallel on the same row
    - **“U” Unite** - Merges two parallel lanes back into a single signal

- **Customizing Menus** - These menus allow the user to change the effect list in their Daisy Chain. The menu buttons are found at the top of the Daisy Chain.
    - **Add an Effect** - Click the “Add” button, and select the effect you would like to add. Will add the specified effect with its default parameters
    - **Copy an Effect** - Click the “Copy” button and select the effect you would like to copy. This creates an exact copy of the effect, keeping its same parameters.
    - **Delete an Effect** - Click the “Del” button and select the effect to be deleted.

5. Effect Modules

After selecting any effect in the Daisy Chain, its parameters and visualizer panel open in the Panel View. Each effect contains its own set of sliders, buttons, and controls that shape the behavior of the effect. Adjusting these parameters updates the sound in real time. The Panel View also displays a matching visualizer beneath the controls.

5.1 Pitch

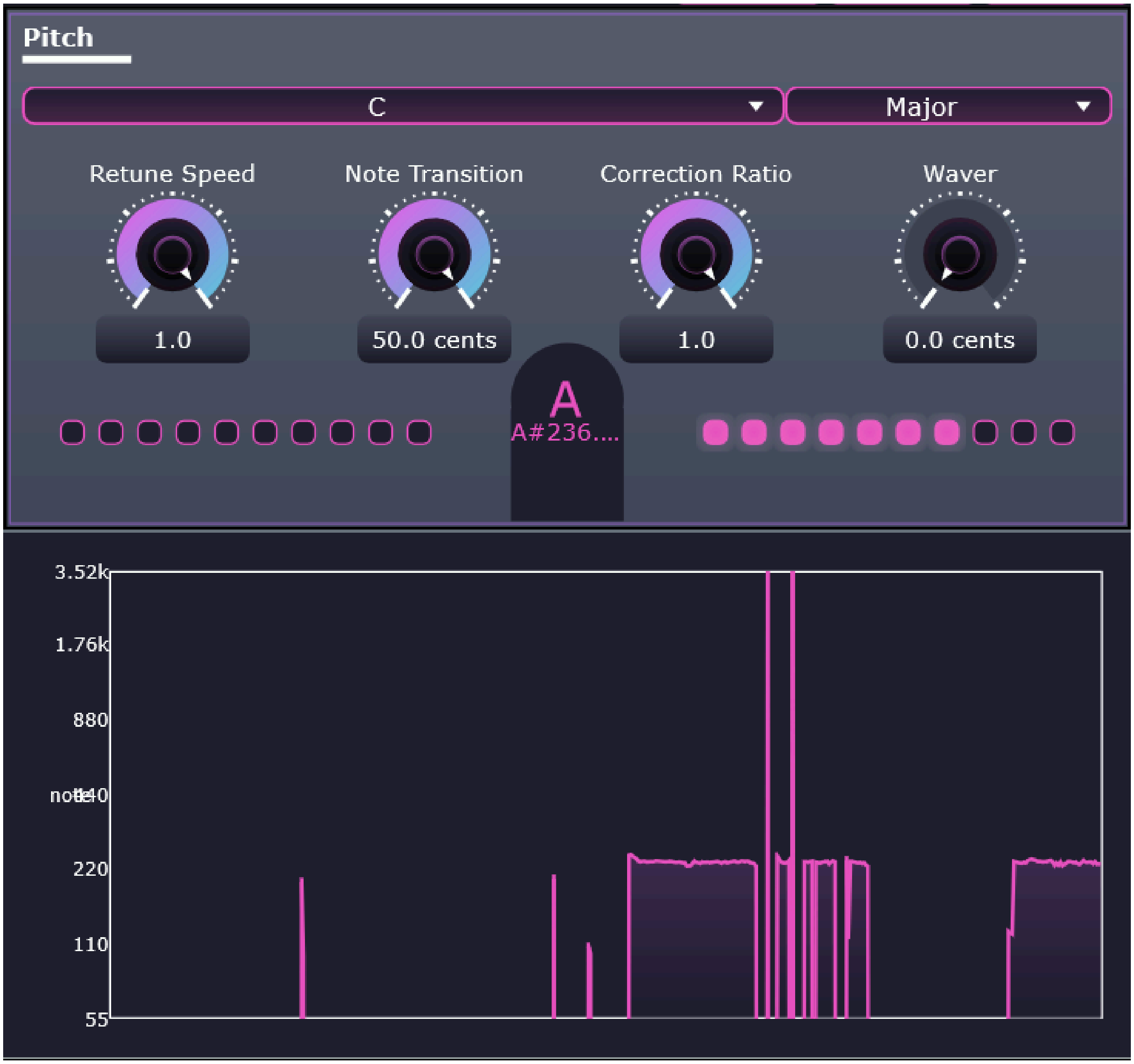
The Pitch module identifies incoming pitch and corrects it to a target pitch according to user-specified parameters.

Parameters:

- Retune Speed - Rate that pitch corrects from natural to target pitch. High values create a robotic Cher effect and low numbers create a natural effect.
- Note Correction - Minimum threshold of pitch change in cents to trigger new target pitch. Low values create fast retune toggles on very sharp notes. High values hold a wobbling pitch on the same note for longer.
- Correction Ratio - Amount of pitch offset to correct by. High values reach perfect correction while low values are closer to the original signal.
- Waver - Amount of pitch modulation to reintroduce after tuning, in cents. Connected to a sinusoid LFO.

Visualizer:

- The parameter panel shows the difference between target pitch and actual pitch.
- The visualizer for this panel shows the detected pitch path over time.



5.2 Formant

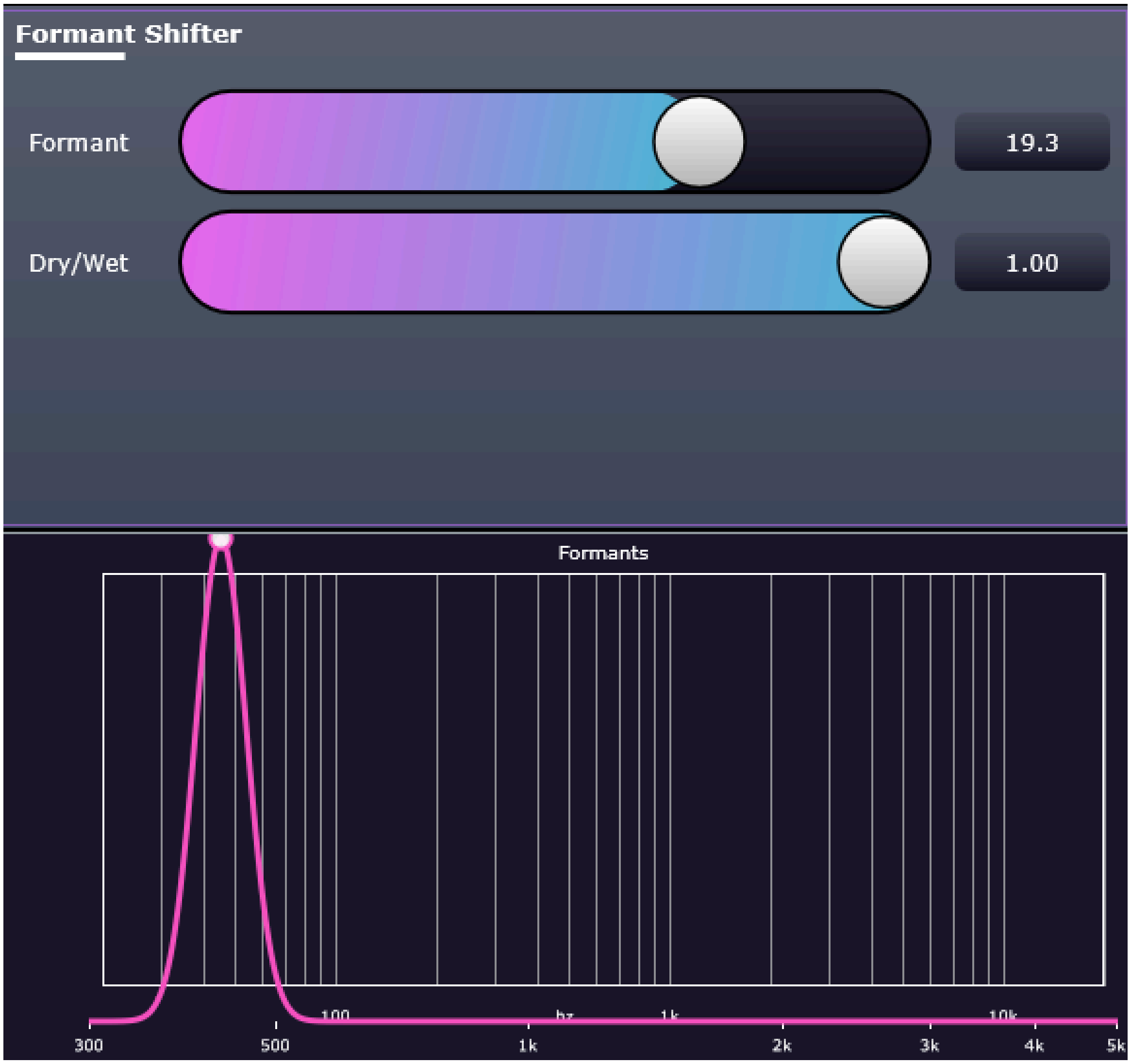
The Formant module changes the timbre and characteristic of the sound.

Parameters:

- Formant - Amount of offset to shift formant by. Shifting upwards makes the timbre more squeaky and pinched, and shifting downwards makes the timbre broader and lower.
- Dry/Wet - Mix between original and altered signal. High value is fully altered, low value is fully bypassed.

Visualizer:

- Identifies the three most prominent formants on the frequency spectrum and shows where they are shifted to.



5.3 Equalizer

The Equalizer module boosts and cuts particular frequencies on a three-band interface.

Parameters:

- Low Freq - Frequency that the low band either cuts or boosts.
- Mid Freq - Frequency that the mid band either cuts or boosts.
- High Freq - Frequency that the high band either cuts or boosts.
- Low Gain - Amount that is boosted or cut at low frequency. Low shelf filter.

- Mid Gain - Amount that is boosted or cut at mid frequency. Band filter.
- High Gain - Amount that is boosted or cut at high frequency. High shelf filter.

Visualizer:

- The parameter panel shows dials to specify the frequencies and gains for each.
- The visualizer shows the gain boosts and cuts, mapped to a grid of decibels by frequency.

5.4 Gain

The Gain module increases or decreases the volume of the audio passing through it.

Parameters:

- Gain - The amount in dB to increase or decrease the volume by.

Visualizer:

- The visualizer for this panel shows the current dB value for the signal.

5.5 Noise Gate

The Noise Gate module silences audio that falls below a certain volume, which is useful for removing background hiss between speaking.

Parameters:

- Threshold - The level at which audio below is muted.
- Attack - How quickly the gate opens when a sound goes above the threshold.
- Release - How quickly the gate closes when a sound goes below the threshold.

Visualizer:

- The visualizer for this panel shows the current dB value for the signal.
- The visualizer for this panel also shows the current threshold (visible as a dotted line).

5.6 Compressor

The Compressor module reduces the dynamic range of audio by making loud sounds quieter.

Parameters:

- Threshold - The level at which audio above is reduce.
- Ratio - How much gain reduction is applied once the threshold is crossed.
- Attack - How quickly the compressor reduces volume.

- Release - How quickly the compressor stops reducing volume.

Visualizer:

- The visualizer for this panel shows the current dB value for the signal.
- The visualizer for this panel also shows the current threshold (visible as a dotted line).

5.7 De-Esser

The De-Esser module reduces harsh “s” and “t” sounds, otherwise known as sibilance.

Parameters:

- Frequency - The specific frequency to target.
- Threshold - The level at which audio above is reduced.

- Ratio - How much gain reduction is applied once the threshold is crossed.
- Attack - How quickly the de-esser reduces volume.
- Release - How quickly the de-esser stops reducing volume.

Visualizer:

- The visualizer for this panel shows the Fourier Transform of the current sound, meaning that it shows the current dB value on the y-axis, and it graphs those over the current frequency value in Hz on the x-axis.
- The visualizer for this panel also shows the current frequency and threshold (visible as vertical and horizontal dotted lines respectively).

5.8 De-Noiser

The De-Noiser module uses spectral subtraction to remove consistent background noise (like fans or hums).

Parameters:

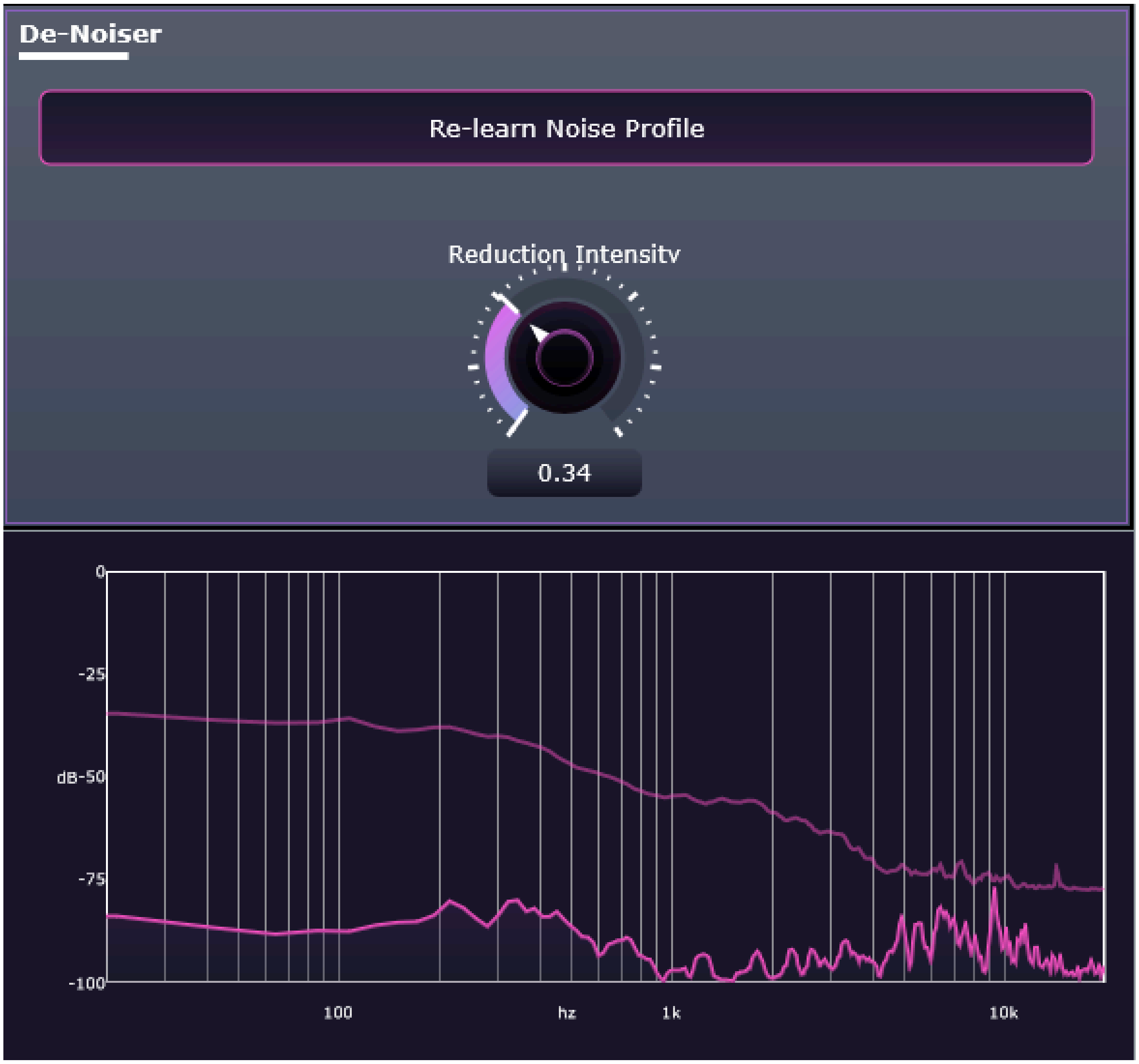
- Learn Button - Allows you to learn or re-learn the noise profile.
- Reduction Intensity - The aggression of the de-noising effect. Setting it too high might degrade the vocal quality while setting it too low might not remove background noise.

Visualizer:

- The visualizer for this panel shows the Fourier Transform of the current sound, meaning that it shows the current dB value on the y-axis, and it graphs those over the current frequency value in Hz on the x-axis.
- The visualizer for this panel also shows secondary information detailing the current captured noise profile, which can be seen at a lower opacity in the background.

How to use:

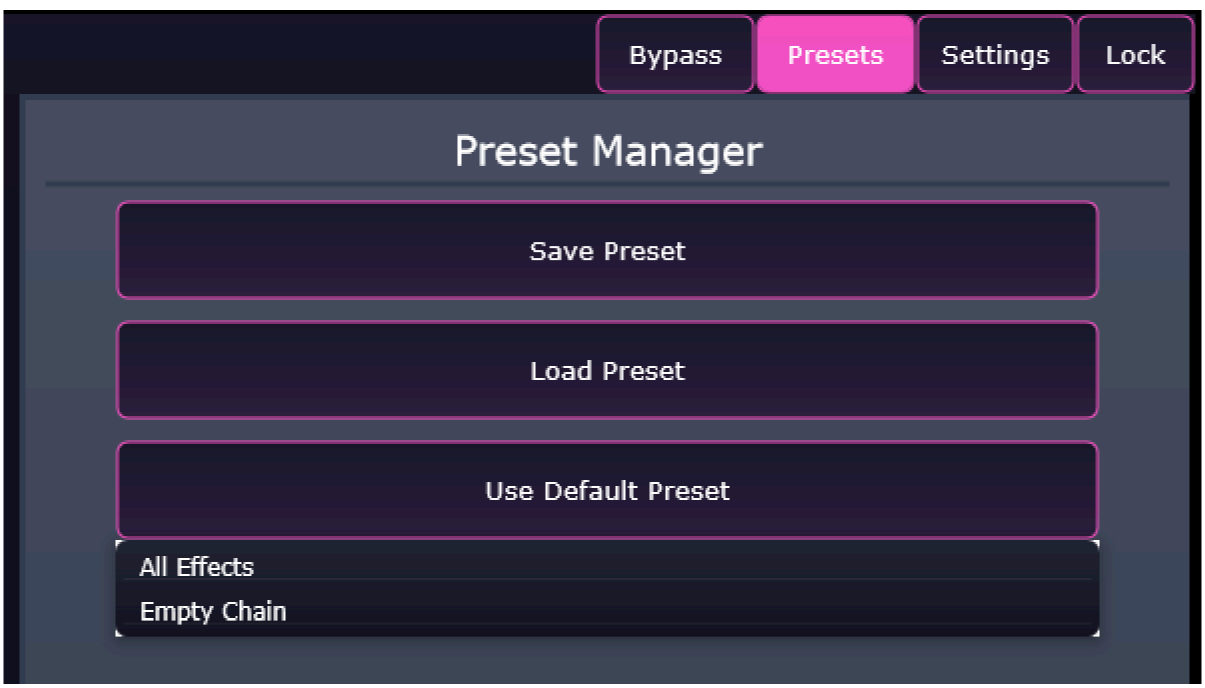
1. Ensure your microphone is active, but do not speak.
2. Click the “Learn” button.
3. Wait 2 seconds.
4. Adjust the “Reduction Intensity” knob until the noise disappears without degrading the voice.


De-Noiser
Re-learn Noise Profile
Reduction Intensity
0.34
0
-25
dB-50
-75
-100
100
hz
1k
10k

6. Preset Management

The Preset Management panel allows you to save your current chain configuration and load previously saved settings. This is useful for recalling specific vocal sounds for different audio or speakers. The Preset Management panel can be accessed by clicking the “Presets” button in the top bar. Once opened, you will be given the following options:

- **Save Preset** - Opens a file dialog to save your current effect chain and parameters as a *.xml file to your computer with the default location of Documents/Pitchblade/Presets/.
- **Load Preset** - Opens a file dialog to load an existing *.xml preset file.
- **Use Default Preset** - Opens a dropdown menu that offers two quick options.
    - **All Effects** - Loads the standard default chain with all effects in default chaining mode: (Gain > Noise Gate > Compressor > De-Esser > De-Noiser > Formant > Pitch > Equalizer).
    - **Empty Chain** - Clears all effects from the daisy chain, giving you a blank slate.



7. Bypass and Lock

You can choose to bypass all effects in two specific ways. This can be useful if you want to rapidly swap between hearing the raw signal and the processed signal for comparison purposes.

- **Global Bypass** - Click the "Bypass" button in the top bar to bypass all effects using the individual bypasses. In this state, you can deselect specific effects to hear them specifically.

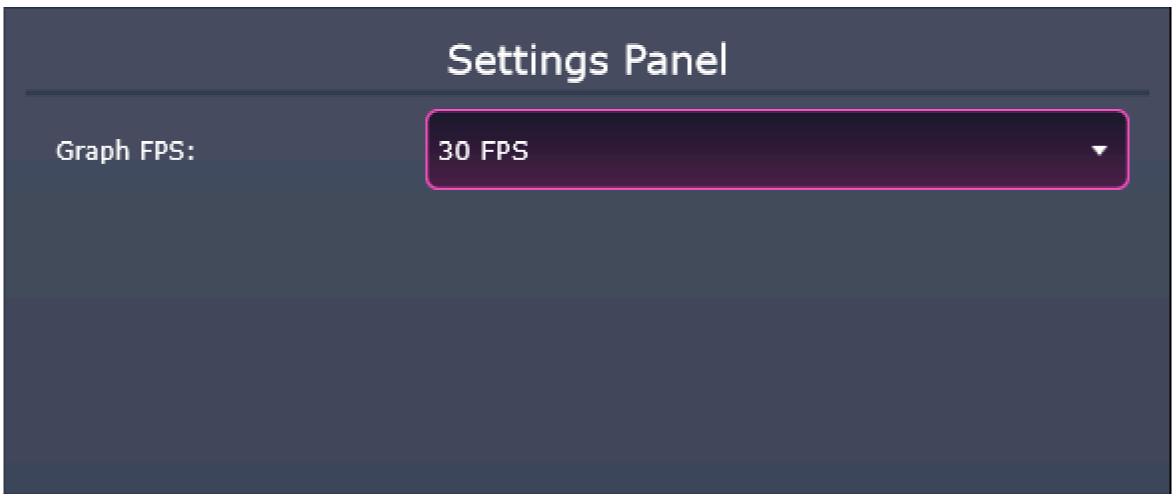
- **Lock** - Click the "Lock" button in the top bar to bypass all effects globally. In this state, the daisy chain is unable to be interacted with. This is useful if you want to preserve which effects were bypassed originally, yet still want to listen to your raw audio.

PITCHBLADE
Bypass
Presets
Settings
Lock
Add
Copy
Del
Gain
D
B
Noise Gate
De-Esser
Compressor
De-Noiser
Formant
Pitch
Equalizer
Gain
0.0 dB
0
-20
-40
dB
-60
-80
-100

8. Settings

The Settings menu can be accessed by clicking the "Settings" button in the top bar. Once opened, you will be able to alter the following:

- **Global Framerate** - By clicking in the dropdown menu, you can select the framerate, or FPS (frames-per-second), that the visualizers print information onto the screen. You have the option to select 5 FPS, 15 FPS, 30 FPS, or 60 FPS. The default framerate for the visualizers is 30 FPS.



9. Troubleshooting

Issue	Common Solution
Plugin not showing in DAW	• Ensure the Pitchblade.vst3 file is located in the correct system folder (C:\Program Files\Common Files\VST3). • Ensure that Pitchblade is built in Release Mode. • Ensure that Pitchblade is compiled from git repository and has access to dependent libraries. • Rescan plugins within your DAW
No Audio Output	• Check Options menu in top left to ensure that correct output device is selected. • Check if the Noise Gate threshold is set too high, which might be cutting off the entire signal. • Check if Gain is turned down significantly. • Check if the Compressor threshold is set too low, which might be reducing the entire signal.
High Latency or Visualizer Lag	• Open the Settings menu and lower the Global Framerate. • Pitch correction requires a small amount of lookahead, so ensure your DAW buffer size is set reasonably.
Distorted Audio	• Check if Gain is turned up significantly. • Check if Pitch and Formant are simultaneously enabled. This is a bug and should be registered as such on the Pitchblade Github.
Robotic Audio	• Check to see if the Pitch retune speed is set too fast, which might be causing the voice to sound robotic.